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Title: Project Data Summary
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Repository: https://github.com/Addie-Shanti-Org/GenderDisparityinResearch

Heard but Not Studied: Mapping the Gap Between Health Research and Women's Lived Experiences

Research Question

Original RQ: What women's health symptoms and conditions are most discussed in online communities, and to what extent are these topics represented in clinical research?

New RQ: Since we've narrowed data sources, our research question becomes:
What women's health symptoms and conditions are most discussed in Reddit subreddits, and to what extent are these topics represented in clinical research on PubMed and ClinicalTrials.gov?

Data Inventory

Reddit Data:
Source: https://academictorrents.com/details/3e3f64dee22dc304cdd2546254ca1f8e8ae542b4
This source contains all reddit posts and comments from 2005-06 to 2025-12, organized by subreddit. Because it is a large dataset (3.97TB), we have to access it via torrent. To do this, we download the torrent file, open it with a torrent client (we used qBittorrent), select which subreddit(s) we want to download, and run the torrent. We downloaded only the r/WomensHealth subreddit data. This lives in `data/raw/subreddits25/` and has both `WomensHealth_comments.zst` (98MB) and `WomensHealth_submissions.zst` (43MB). We unpacked the submission's zst file and put the contents in `data/interim/subreddits/WomensHealth/`, only a JSON file called `submissions` (370MB). This JSON file was unpacked to a dataframe and after some exploratory analysis (code in `notebooks/Reddit_Data/`), we uploaded it to the PostgreSQL database on Railway in a python notebook, `notebooks/Reddit_Data/raw_to_db.ipynb`. In this notebook, we uploaded the whole dataset to a table named `raw_submissions`, which has 123997 rows and 128 columns. We picked columns that we thought would serve analysis the best while reducing the amount of data processed and made a new table, `submissions` with columns `['id', 'title', 'selftext', 'upvote_ratio', 'ups', 'num_comments',`

`'subreddit']`. This data doesn't need to be updated, and if we want data from other subreddits we will follow the same procedure.
We've started doing some text analysis in `notebooks/text_analysis.ipynb`. The results of the text analysis lives in `data/processed/`.

PubMed Data
Source: paperscraper.pubmed library in python
(https://jannisborn.github.io/paperscraper/api/pubmed/
https://github.com/jannisborn/paperscraper)
License/Terms:
paperscraper is maintained as an open-source research software project. Its scope is to provide practical tools for scholarly metadata collection, publication search, full-text retrieval where access is available, citation analysis, and related reproducible examples.

ClinicalTrials Data
Source: pytrials library in Python
(https://pypi.org/project/pytrials/
https://github.com/jvfe/pytrials
https://clinicaltrials.gov/data-api/about-api)

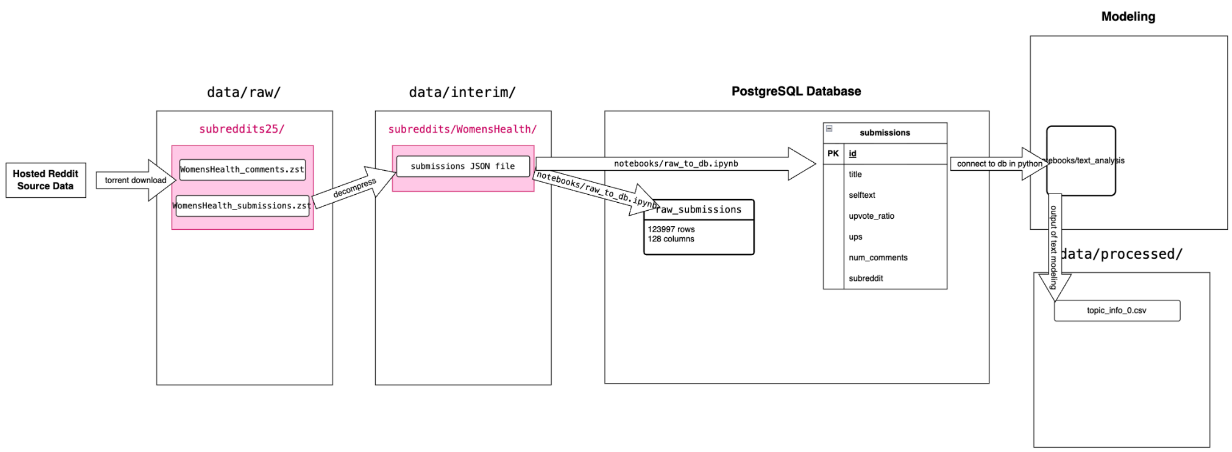
License/Terms:
Free software under the BSD license.

PubMed and ClinicalTrials API Query
These two APIs were queried by filtering for keywords that can be found here:
https://www.brighamandwomens.org/womens-health/diseases-conditions
This was done to capture topics using medical terminology that is most likely not used in reddit submissions. These conditions and diseases are what the Women's Health Center of Excellence of Mass General Brigham researches to provide "...specialty medical care for women throughout their health journey at all life stages." The dates that were chosen for the queries were based on the first publications of any keyword in our topics list from the source above.

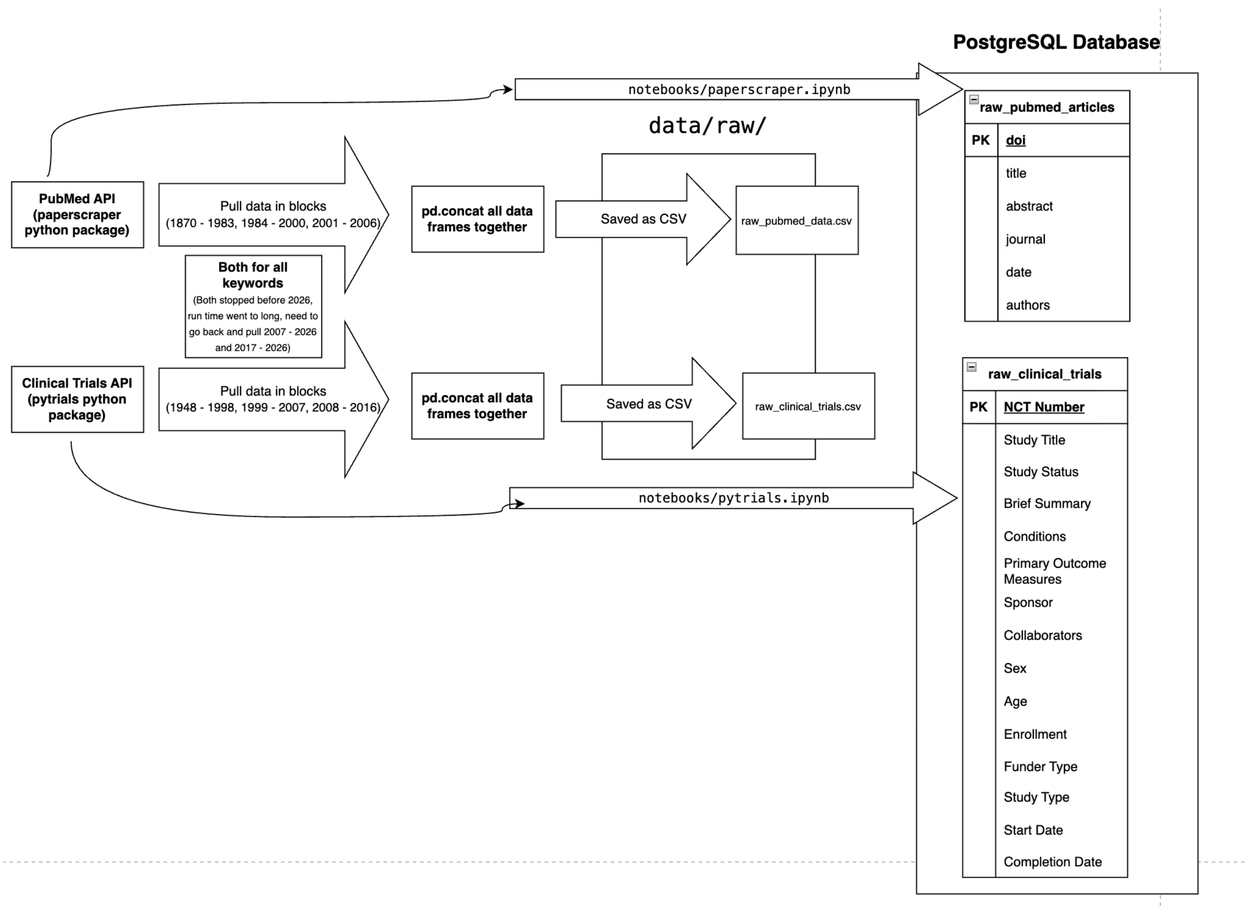
Data Source	Row and Column Counts	Date Range	Refresh Policy	Location
Reddit Data **Table:** submissions	Rows: 123997 Columns: 7	2005-06 - 2025-12	N/A	data/processed/
PubMed API **Table:** raw_pubmed_articles	Rows: 2,184,056 Columns: 6	1870 - 2006	Need to do manual pull for current dates	data/raw/
ClinicalTrials API **Table:** raw_clinical_trials	Rows: 255,783 Columns: 15	1948 - 2016	Need to manual pull for current dates	data/raw/

Schema and Organization

Reddit Data Raw → Processed



PubMed and Clinical Trials Data Raw (Need to add rest of dates and query for reddit data analysis)



Ingestion Status

Reddit Data:
Manual pull for r/WomensHealth is completed and no further action is required. However, if we want to ingest other subreddits, we still have access to this data and can pull these in the same manner.

Pubmed Data:
Manual pull for the pubmed_data is completed for years 1870 - 2006. Since there is so much data including years and query keywords, and how the data was being pulled, by month for each keyword between 2001 - 2026, the script was taking too long. We will still need to pull for 2007 - 2026. Although, we do have quite a bit of historical data that we can work with and do frequency analysis on.

Clinical Trials Data:
Manual pull for the clinical_trials_data is completed for years 1948 - 2016. This was also having the same issues as the pubmed data, with the data being pulled the same way. We will still need to pull for 2017 - 2026.

Data Quality and Profiling

Reddit data:
Exploratory analysis has been done in `notebooks/explore_raw_a.ipynb. The reddit data has no duplicates. However, many columns are completely empty, so those will be dropped. Many reddit titles say [deleted by user] or [removed by moderator]. This will mess up the text analysis, so those will be dropped. A first round of text analysis was done on the raw data, and it exposed the redundant, unrelated terms used. The data we want to focus on is text, so not much outlier detection can be done.

PubMed Data:
Duplicate checking was done before loading the raw combined csv into the warehouse. Duplicates were checked on DOI and title, as DOIs were introduced in 1997, so earlier papers might not have a unique DOI. Total rows went from 3,166,600 to 2,184,056.
PubMed does not use DOI to uniquely identify a paper, they use PMID and PMCID, which paperscraper does not retrieve (https://library.stevens.edu/c.php?g=442331&p=6577176).

Columns with NAs →

Currently there are about 665,223 rows without a DOI, we will need to figure out a way to join PMID to our dataframe, once we have all our dates. Here is the converter:https://pmc.ncbi.nlm.nih.gov/tools/idconv/

The abstract also has 477, 289 NAs. paperscraper has a hard time pulling abstract because of API restrictions. They also did not index abstracts before 1975.

Columns with no NAs →
Title, Date, and Authors

Clinical Trials:
Duplicate checking was done after loading the raw data into the warehouse, in ct_explore_raw_data.ipynb in notebooks/Article_Data_APIs. Duplicates were filtered on NCT Number, which is the primary key or unique identifier for this dataset.
Columns with no NAs → NCT Number, Study Title, Study Status, Brief Summary, Conditions, Sponsor, Age, Funder Type, Study Type, and Start Date.
Columns with NAs → Primary Outcome Measures (9521), Collaborators (96141), Sex (98), Enrollment (3069), and Completion Date (7942).

Column names were renamed in notebooks/Article_Data_APIs/ct_explore_raw_data.ipynb to be consistent with lower case, snake case structure.

Reproducibility

Our notebooks have the necessary `pip install` and import commands built in for any packages we need. We have `.env` files set up to hold database connection info.

Ethics and Access Controls

Reddit Data:
We drop the username column entirely before moving to our analysis step. Further work is to be done to ensure no sensitive data is in the text and titles.

PubMed and Clinical Trials:
Since all of this data is from open APIs, there is no PII handling or redaction needed on our part, this is due to the restrictions put on the API by the sites themselves.

Freeze Statement

Our sources are confirmed and we don’t have any further work on exploring and testing sources. Due to the nature of our work, we will have cycles of ingestion and analysis being done, including adding current date data for PubMed and Clinical Trials datasets.